

Automation Risk and the Public Balance Sheet

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Abstract

Workers can be exposed to automation states that they cannot trade, while the assets available to the government pay in only some of those states. This paper asks when a public risky holding is insurance for workers rather than simply additional market exposure on the public balance sheet. In a small open economy with hand-to-mouth workers, privately owned domestic capital, and gradual capital-tax adjustment, I price the same public payoff in two ways: by the international investors who sell it and by the government that values the worker resources it delivers. At an interior Ramsey allocation, these valuations agree along the traded payoff. They need not agree in an orthogonal direction. A small-risk decomposition separates the public position into ordinary compensated-return demand and a hedge against fiscal-resource risk; the remaining projection error determines the leading welfare value of an additional payoff. Capital taxation cannot reproduce this impact transfer because it changes capital and fiscal resources through time. The same logic applies to a transformative arrival with two possible successor economies: one inherited equity payoff supplies one transfer direction across them, not two. The relevant policy statistic is therefore the alignment of worker-resource exposure with the public payoff menu, not the asset's label, expected return, or gross holding.

Keywords: automation, public portfolios, capital taxation, incomplete markets, fiscal insurance.

JEL codes: E62, G11, H21, H63, O33.

1 Introduction

Suppose an automation innovation raises output and the return to capital but reduces the resources reaching workers on impact. Public equity seems to offer an immediate answer: buy the asset that gains from automation and transfer the proceeds. Now change the innovation so that workers are still exposed but the same equity claim does not move. Buying more of it cannot help. The difference between these cases is not the average return on equity, the severity of automation, or the size of the public fund. It is the state-contingent alignment between what workers need and what the asset pays.

This paper studies that alignment. The government acts for workers who do not trade, but it buys financial claims at prices set by international investors. An extra public dollar is valuable when it supports worker consumption, wages, or future tax capacity. The same dollar is expensive to acquire when it pays in states that market investors also value highly. A public risky holding is worker insurance only when its payoff loads on the difference between those two valuations. Public trade can remove that difference along the claim's payoff. If worker resources also vary in another direction, the residual is not repaired by holding more of the same claim.

The model combines a task-based production economy with a public balance sheet. Two continuous shocks move ordinary productivity and the range of tasks that capital can perform. Workers

consume their wage plus transfers and have no asset market access. Domestic capitalists own installed capital and choose accumulation. The government chooses worker consumption, a safe position, one internationally traded risky claim, and the speed at which a source-based capital-income tax changes. The risky claim moves public resources across the states in its payoff. Safe finance moves them across dates. The tax moves resources from domestic capital income toward workers but changes the return that sustains future capital, wages, and the tax base. These are distinct instruments even when each ultimately finances the same worker consumption.

The main result prices a marginal public holding twice. Let σ_R denote the claim's Brownian payoff loading, and let λ_I and λ_G denote the market and fiscal prices of risk. The derivative of the Ramsey Hamiltonian with respect to the dollar holding s satisfies

$$\frac{H_s(s)}{V_N} = \sigma_R^T [\lambda_I - \lambda_G(s)]. \quad (1.1)$$

At an unconstrained interior choice, the right-hand side is zero. The conclusion is a projection, not equality of the two valuation kernels. It says that the government exhausts gains from trade in the direction it can buy; it says nothing comparable about directions outside the asset menu.

A small-risk expansion translates this valuation result into a portfolio and welfare decomposition. Let J be the value of future wages and tax resources under the deterministic policy problem, and let $X = N + J$ combine it with public financial net worth N . If h_J is the claim-equivalent projection of fiscal-resource risk, the leading risky holding is

$$s^0 = X - h_J. \quad (1.2)$$

The first term is ordinary log-portfolio demand for compensated market exposure. The second is a fiscal hedge. Their sum is not an insurance statistic: a large public long position can be driven mainly by compensated return demand, while an important worker exposure remains orthogonal to the claim. At order ε^2 , the welfare value of a second payoff is proportional to the squared orthogonal exposure. This comparison makes clear why a payoff with no additional expected excess return can nevertheless be valuable.

Automation supplies unusually sharp versions of both alignment and failure. At inherited capital, one shock direction holds aggregate output fixed while reallocating income from labour toward capital; taxation recovers only part of that loss for workers. A different direction leaves the traded claim unchanged while worker and fiscal resources fall. The first case is potentially insurable through the claim. The second is a negative control: it is economically important automation risk that the existing payoff does not reach. Figure 1 introduces these experiments before the recursive planner problem.

Taxes respond on another clock. The risky payoff reaches public net worth with the shock, whereas installed capital and the statutory tax rate are inherited. Changing the tax affects future resources through current redistribution and the subsequent capital path. The optimal tax-speed condition therefore depends on the entire discounted path of the tax base and on the fiscal value of preserving capital, not merely on the impact loss in wages. A capital tax can alter the exposure that the government eventually bears; it does not create a missing event payoff at the instant of the shock.

The final application changes the timing but not the question. Before a transformative arrival, the future economy may remain labour using or become a labour-free AK economy. One inherited equity position delivers a vector of payoffs across those two successors. Changing the scalar position moves successor public wealth along that vector; it does not choose wealth in the two economies

independently. This is an exact rank statement for the named equity-and-flow-tax menu. It is not, by itself, a welfare ranking or an implementation theorem.

The closest conceptual precedent is Aparisi de Lannoy et al. (2025), who place relative stochastic discount factors inside public-portfolio choice and separate distributional hedging from other portfolio motives. I inherit that valuation logic and ask a narrower question: which automation exposures are reachable when two economically relevant risks confront one public payoff? This also puts the paper in the incomplete-market Ramsey tradition. Farhi (2010) and Bhandari et al. (2017) show how restricted security menus and public ownership change fiscal implementation. Here the restriction is tied to worker exposure under automation, and completing the public payoff span is kept distinct from completing household markets or reaching a first best.

The task-automation literature provides the incidence states. Productivity, displacement, and capital accumulation can move output, wages, and capital returns differently (Acemoglu and Restrepo, 2019; Moll et al., 2022). Recent work on transformative AI also makes clear that partial automation, weak-link economies, and scalable-capital limits imply different paths for wages and labour’s share (Aghion et al., 2019; Korinek and Suh, 2024; Restrepo, 2025; Jones and Tonetti, 2026). Korinek and Lockwood (2026) already proposes public ownership as preparation for this uncertainty. The contribution here is not the broad idea of public equity insurance. It is the payoff test for when that ownership transfers resources to the worker states for which it is intended, and the characterization of what remains when it does not.

The analysis uses worker continuation welfare in a price-taking small open economy. The formal results condition on the stated smooth production branch and the relevant interior controls; the welfare formulas based on fiscal-resource projections are local to small risk. These restrictions preserve the paper’s question: given a public asset menu, which worker exposures can it intermediate? They do not turn the answer into a calibrated portfolio recommendation.

The rest of the paper proceeds from the state-payoff experiment to the model and then to the valuation result. Section 2 closes the economy and shows how automation moves worker resources and claim payoffs. Section 3 derives the two valuations of a public payoff. Section 4 connects that result to the leading holding, residual exposure, and welfare value of another payoff. Section 5 explains why taxes work through the transition rather than as an impact hedge. Section 6 applies the same payoff-menu criterion to two possible successor economies.

2 The economy behind the insurance problem

The question is not whether an asset has a high expected return, or whether an automation shock is bad in the aggregate. It is whether the public balance sheet can move resources into states in which workers are relatively exposed. Workers cannot save or trade; domestic owners choose and own domestic capital; an international market prices the claim that the government may buy or sell. The government can transfer current resources to workers, carry a safe position across dates, take a position in that external claim, and change the source-based tax on domestic rental income only gradually.

Sequential closure. At each date, competitive firms rent the predetermined domestic capital stock and hire labour on the maintained full-specialization branch. Workers are hand-to-mouth: they consume wages plus transfers and neither save, borrow, nor trade. Domestic capitalists own installed domestic capital, consume and invest out of its after-tax rental income, and do not sell that domestic productive claim into the international market. The government chooses transfers—

equivalently, worker consumption subject to the transfer floor—a dollar position in an *external risky claim*, and tax speed. Foreign investors take residual positions and price the safe and external risky claims from the given world price block. Goods and public-budget identities then clear, and the Ramsey problem chooses policy over the induced equilibrium. Thus the public risky position is not public ownership of domestic installed capital, and workers do not privately finance it.

2.1 States, production, and ownership

Two independent Brownian innovations, $dW_t = (dW_t^P, dW_t^A)^\top$, move ordinary productivity z_t and the latent automation state x_t :

$$dz_t = \kappa_z(\bar{z} - z_t)dt + \sigma_z dW_t^P, \quad dx_t = \kappa_x(\bar{x} - x_t)dt + \sigma_x dW_t^A, \quad d\langle W^P, W^A \rangle_t = 0. \quad (2.1)$$

The available automation-intensive task share is $\alpha_t = A(x_t)$, with $A'(x) > 0$. “Impact” below therefore means a local innovation at inherited capital, not a discrete jump.

A competitive domestic firm allocates installed capital K_t and a fixed labour supply L across automation-intensive and new tasks. On the maintained full-specialization branch, output and factor prices are

$$Y = e^{z+\nu_A\alpha}\Omega(\alpha)K^\alpha L^{1-\alpha}, \quad R^K = \alpha \frac{Y}{K}, \quad w = (1-\alpha)\frac{Y}{L}, \quad (2.2)$$

where $\Omega(\alpha) = \alpha^{-\alpha}(1-\alpha)^{-(1-\alpha)}$. The displayed formula holds only while the strict task-specialization inequalities hold; the global firm problem selects the allocation at their boundaries.

Workers supply L , own no financial assets, and consume their wage bill plus a nonnegative public transfer:

$$c_t = w_t L + T_t, \quad T_t \geq 0. \quad (2.3)$$

Domestic capitalists own all installed domestic capital. With reversible installation and log preferences, their capital law on this branch is

$$dK_t = K_t[(1-\tau_t)(R_t^K - \delta) - \rho]dt. \quad (2.4)$$

The government does not choose investment directly. It changes the after-tax return through the source tax τ_t on signed net rental income $\mathcal{B}_t = (R_t^K - \delta)K_t$. The tax rate is inherited and moves only at speed ν_t , $d\tau_t = \nu_t dt$, at resource cost $Y_t \nu_t^2 / (2\kappa_\tau)$. Capital and the tax rate therefore cannot deliver a contemporaneous Brownian payoff; they change the subsequent path of wages, the tax base, and capital.

2.2 Prices and the public balance sheet

The domestic economy is small in world financial markets. An exogenous international stochastic discount factor prices a safe asset and one external risky claim. Writing $\lambda_{I,t}$ for the market price of Brownian risk, the claim’s total return is

$$d\mathcal{R}_{R,t} = (r_{I,t} + \beta_{R,t})dt + \sigma_{R,t}^\top dW_t, \quad \beta_{R,t} = \sigma_{R,t}^\top \lambda_{I,t}. \quad (2.5)$$

Expected excess return compensates exposure to states valued by the marginal international investor. It does not by itself say that the claim insures domestic workers. In the maintained specialized world closure $\sigma_R = \lambda_I$, but the argument below requires only the stated payoff loading.

Table 1: State-contingent worker resources and the maintained claim

Impact experiment	Worker and fiscal resources	Claim payoff	Insurance implication
Output-neutral redistribution, $dz = -s_A d\alpha$	$dY = 0$, $dW = -Y d\alpha$, and $d\mathcal{F} = -(1 - \tau)Y d\alpha$ at inherited capital.	Positive under the stated domestic– international alignment	The claim can move public resources toward the worker-loss state, subject to its price and the government’s constraints.
Claim-neutral automation, $dz = -h_I d\alpha$	$dY = dW = d\mathcal{F} =$ $-Y d\alpha/\alpha$ at inherited capital.	Zero on impact under the same alignment	A material worker and fiscal loss lies outside this claim’s payoff direction.

Notes: The first two rows are local analytical experiments on the full-specialization branch at inherited capital. A wage change is not automatically a consumption change because transfers may respond. The table reports no calibrated magnitude or complete transition.

Let N_t denote public financial net worth, measured in goods. The government holds s_t dollars of the external risky claim; its residual safe position is $N_t - s_t$. Consolidating the worker and public accounts is useful for planning but does not turn wages into Treasury revenue. Define current worker-planner resources before consumption and tax-adjustment costs by

$$\mathcal{F}_t = \tau_t \mathcal{B}_t + w_t L. \quad (2.6)$$

The public balance sheet then evolves as

$$dN_t = \left[r_{I,t} N_t + s_t \beta_{R,t} + \mathcal{F}_t - c_t - \frac{Y_t}{2\kappa_\tau} \nu_t^2 \right] dt + s_t \sigma_{R,t}^\top dW_t, \quad (2.7)$$

$$\mathcal{F}_t - c_t = \tau_t \mathcal{B}_t - T_t. \quad (2.8)$$

The second line restores the Treasury account: wages finance worker consumption directly and cancel from the consolidated flow. The first line isolates the timing distinction that will matter throughout the paper. A pre-existing risky position moves public wealth with the innovation along σ_R ; safe finance moves resources across dates; and tax adjustment moves the future domestic return and hence the future tax base and wage bill. Wage income is private and nonpledgeable; future worker resources can matter for welfare without securing public debt.

Figure 1 translates the first two rows of Table 1 into the central experiment. In the first, the claim pays in the state in which current worker resources fall; it can be insurance, subject to its price and the government’s constraints. In the second, the same worker loss has zero impact exposure to the maintained claim. Calling the asset “automation-linked” does not repair the missing payoff.

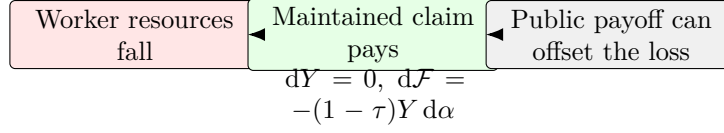
2.3 Equilibrium and the Ramsey insurance problem

For a given admissible policy, a competitive equilibrium consists of the firm’s task allocation and factor prices, domestic owners’ capital accumulation, worker consumption in (2.3), international prices in (2.5), and a public balance sheet satisfying (2.7). Goods, domestic capital, and the stated international claims clear under the small-open closure. Workers do not acquire a hidden private hedge: all insurance they receive passes through wages, transfers, and the public balance sheet.

At date zero the government commits to (c_t, s_t, ν_t) to maximize

$$V(S_0) = \sup_{(c,s,\nu) \in \mathcal{A}(S_0)} \mathbb{E}_{S_0} \int_0^\infty e^{-\rho t} \log c_t dt, \quad S_t = (z_t, x_t, K_t, N_t, \tau_t). \quad (2.9)$$

Output-neutral redistribution



Claim-neutral automation

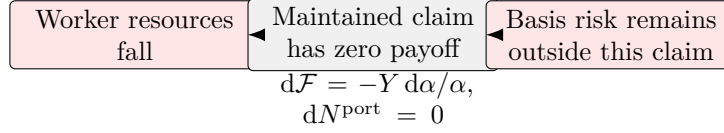


Figure 1: When the external risky claim is and is not worker insurance. The diagram is a normalized rendering of the two local experiments in Table 1. Insurance depends on the alignment of the claim payoff with worker-resource losses, not on the asset label or expected return.

Table 2: The policy margins are distinct

Instrument	Direct object moved	Margin it cannot replace
Safe position $N - s$	Public resources across dates at the world safe rate	State-contingent transfer across Brownian payoff directions
External risky position s	Public wealth on impact along σ_R	The component of worker-resource risk orthogonal to σ_R
Source-tax speed ν	Future domestic after-tax return, investment, wages, and tax base	The payoff on the Brownian innovation already realized
Transfer T	Current worker consumption, subject to the public budget and $T \geq 0$	Resources absent from the current public balance sheet

The inherited augmented state is (S_0, M_0) ; M_0 is empty in the maintained baseline. Admissibility includes the transfer floor, portfolio and safe-debt bounds, the tax ceiling, continuation solvency, and the usual integrability and transversality restrictions. The recursive representation is a conditional characterization on this named admissible continuation class.

The rest of the paper asks which component of the worker-resource disturbance can be traded through σ_R . The next section answers by comparing the market value of the payoff with its fiscal value inside the Ramsey problem.

3 When equity is insurance

The government buys the risky claim at a market price, but values its payoff by what the resulting public resources do for workers. This section puts those two valuations on the same line. The derivation is deliberately stated in dollar holdings and payoff loadings; the normalized five-state HJB is not needed to see the result.

3.1 One payoff, priced twice

Let s_t be the market value of the public risky holding and let its total return have excess drift $\beta_{R,t}$ and Brownian loading $\sigma_{R,t} \in \mathbb{R}^2$:

$$d\mathcal{R}_{R,t} = (r_t + \beta_{R,t}) dt + \sigma_{R,t}^\top dW_t. \quad (3.1)$$

No arbitrage in the international reference economy gives

$$\underbrace{\beta_R}_{\text{market compensation}} = \underbrace{\sigma_R^\top \lambda_I}_{\text{market value of the payoff exposure}}, \quad (3.2)$$

where the market stochastic discount factor has diffusion $dm_I/m_I = -r dt - \lambda_I^\top dW$.

The same holding enters public net worth as

$$dN = [rN + s\beta_R + \mathcal{F} - c - C^\tau(\nu)] dt + s\sigma_R^\top dW, \quad (3.3)$$

where $\mathcal{F}$ is the flow of wages and tax resources relevant for worker consumption and $C^\tau(\nu)$ is the resource cost of moving the tax. Let V be the continuation value of the Ramsey plan and let $V_N > 0$ and $V_{NN} < 0$ on the smooth interior. If ζ_V is the Brownian loading of V_N generated by the exogenous state shocks before the current portfolio exposure is added, the part of the Hamiltonian that depends on s is

$$H^s(s) = s\beta_R V_N + s\sigma_R^\top \zeta_V + \frac{1}{2}s^2 \|\sigma_R\|^2 V_{NN}. \quad (3.4)$$

The three terms have different jobs. The first values the compensated excess drift. The second records how the payoff covaries with the state dependence of a public dollar. The third is the curvature cost of adding risk to public net worth.

Define the fiscal stochastic discount factor by

$$m_G(t) = e^{-\rho t} V_N(S_t). \quad (3.5)$$

It is the shadow value of a marginal unit of public financial resources along the worker-welfare Ramsey plan. Applying Itô's formula to V_N gives its price of risk:

$$\sigma_R^\top \lambda_G(s) = -\frac{\sigma_R^\top \zeta_V + s \|\sigma_R\|^2 V_{NN}}{V_N}. \quad (3.6)$$

Equation (3.2) values the claim at the margin for its seller; equation (3.6) values the same exposure inside the public problem. Subtracting them is the decisive step.

Proposition 3.1 (Fiscal and market valuation of the traded payoff). *Suppose $V_N > 0$ and $V_{NN} < 0$, $V_N(S_t)$ admits the Itô decomposition used in (3.6), and the predictable portfolio choice is admissible and admits local feasible variations. If the transfer and portfolio choices are interior at the current state, then, for any candidate dollar holding s ,*

$$\boxed{\frac{H_s(s)}{V_N} = \sigma_R^\top [\lambda_I - \lambda_G(s)]}. \quad (3.7)$$

The unconstrained optimum satisfies

$$\sigma_R^\top (\lambda_I - \lambda_G) = 0. \quad (3.8)$$

At an upper portfolio bound the left-hand side is weakly positive as a derivative with respect to relaxing the bound; at a lower bound the corresponding inequality reverses.

The proof is already visible in the two pricing equations. Differentiating (3.4) gives

$$\frac{H_s(s)}{V_N} = \beta_R + \frac{\sigma_R^\top \zeta_V + s \|\sigma_R\|^2 V_{NN}}{V_N}.$$

Substituting (3.2) and (3.6) yields (3.7). Appendix B derives the fiscal SDF from the full state process, records the admissibility conditions, and gives the KKT form at both bounds.

The result says exactly what public trade can accomplish. If the claim pays in states where a marginal public dollar is more valuable than its market price, the government wants a larger holding. Its purchase changes both expected resources and the exposure of public net worth until the valuation disagreement has no component in the payoff direction. Nothing in the first-order condition forces $\lambda_I = \lambda_G$ as vectors. With two shocks and one claim, one component of the disagreement can remain.

3.2 The worker-consumption interpretation

On the transfer interior, the consumption first-order condition is $V_N = 1/c$. If the international market SDF has the familiar log-consumption representation with the same discount rate, the relative fiscal SDF is

$$\Theta_t := \frac{m_{G,t}}{m_{I,t}} = \frac{C_t^I}{c_t}. \quad (3.9)$$

Thus Θ rises when a public dollar becomes more valuable for workers relative to the investor who prices the claim. Condition (3.8) can be written as

$$\frac{\text{Cov}_t[\text{d} \log(c_t/C_t^I), \text{d}\mathcal{R}_{R,t}]}{\text{d}t} = 0. \quad (3.10)$$

The government is not minimizing the variance of worker consumption. It is choosing the marketed state transfer until worker consumption relative to market-investor consumption is orthogonal to the traded payoff.

This distinction handles cases that the phrase “equity insurance” obscures. Equity can insure worker downside when it pays as worker resources fall. It can also share unequal upside when workers gain but investors gain more, so a public dollar is still relatively valuable. Conversely, equity can pay in an automation state without being a worker hedge if workers already do better in relative marginal-value terms. The sign is determined by relative valuation along the actual payoff, not by an unconditional correlation with an automation index.

Figure 2 shows the vector content of the result. Public trade removes the projection of the valuation difference on σ_R . A second independent payoff, not a larger position in the first, is required to reach the orthogonal component. The next section expresses the same geometry in fiscal-resource units and attaches a local welfare value to the missing direction.

4 The missing hedge

Proposition 3.1 is exact, but its fiscal price of risk contains the entire continuation problem. A small-risk expansion separates two objects that are easy to confuse: the size of the public holding and the worker-resource exposure left outside the payoff span.

4.1 From the valuation wedge to a portfolio

Scale the two primitive volatilities by ε . On the maintained world-pricing branch, where the claim loading equals the market price of risk, write

$$\sigma_R = \varepsilon \hat{\lambda}, \quad \beta_R = \varepsilon^2 \hat{\beta}, \quad \hat{\beta} = \hat{\lambda}^\top \hat{\lambda} > 0. \quad (4.1)$$

At $\varepsilon = 0$, let $J(z, x, K, \tau)$ be the value of future worker wages and tax resources under the deterministic policy problem. Define comprehensive worker-planner resources and their state exposure by

$$X = N + J, \quad \zeta_J = \begin{pmatrix} \hat{\sigma}_z J_z \\ \hat{\sigma}_x J_x \end{pmatrix}. \quad (4.2)$$

The notation is economic. N is tradable public financial wealth. J is the value of future resources available for worker consumption. Their sum governs consumption demand on the separated deterministic branch, but J is not a pledgeable security or a debt limit.

The order- ε^2 portfolio first-order condition is

$$\underbrace{\hat{\beta} \left(1 - \frac{s}{X}\right)}_{\text{compensated-return demand}} - \underbrace{\frac{\zeta_J^\top \hat{\lambda}}{X}}_{\text{fiscal-resource exposure}} = 0. \quad (4.3)$$

The first term is the familiar log-portfolio trade-off between the claim's premium and the curvature cost of its risk. The second asks how much of the government's future worker-resource exposure the claim already replicates.

Corollary 4.1 (Return demand and the fiscal hedge). *Suppose the value and optimizer admit the regular small-risk expansion described in Appendix D along a feasible deterministic separated branch, with $X > 0$ and an interior leading portfolio. Define the claim-equivalent fiscal exposure*

$$h_J := \frac{\zeta_J^\top \hat{\lambda}}{\hat{\beta}}. \quad (4.4)$$

Then the leading dollar holding and its residual safe financing are

$$\boxed{s^0 = X - h_J, \quad s^0 - N = J - h_J.} \quad (4.5)$$

The decomposition answers two different questions. X is how much exposure a log-utility investor would demand when the claim's expected excess return equals its variance. The term $-h_J$ changes that position because future worker resources already load on the same payoff direction. A large long position may therefore reflect a large compensated-return component rather than a large insurance benefit. Likewise, $s^0 - N > 0$ describes desired safe borrowing on the unconstrained branch; it does not say that future wages can be pledged to support that borrowing.

4.2 More equity is not another payoff

Let

$$P := \frac{\hat{\lambda} \hat{\lambda}^\top}{\hat{\beta}} \quad (4.6)$$

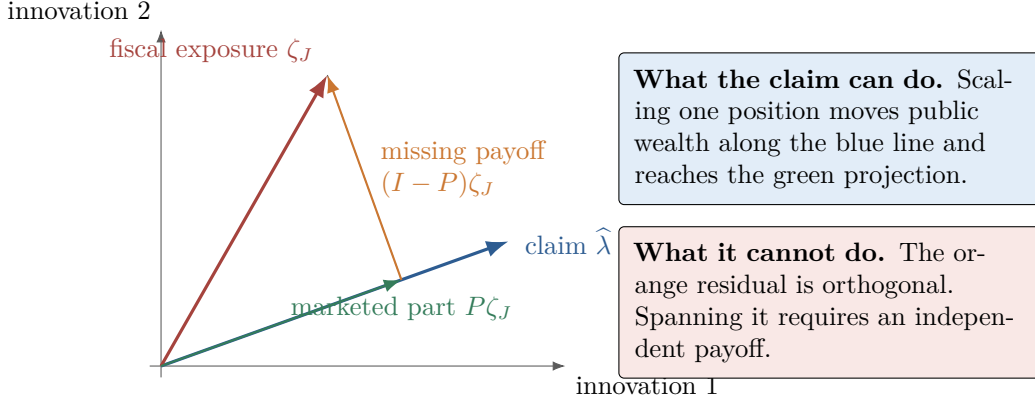


Figure 2: More of the claim versus a different claim. Public trade changes exposure along the maintained payoff and therefore removes only the projection of fiscal-resource risk on that direction. The orthogonal residual requires an independent payoff. The coordinates are normalized analytical quantities, not calibrated magnitudes.

be the projection onto the claim's payoff direction. Fiscal-resource risk decomposes as

$$\underbrace{\zeta_J}_{\text{worker-resource exposure}} = \underbrace{P\zeta_J}_{h_J \hat{\lambda}} + \underbrace{(I - P)\zeta_J}_{\zeta_J^\perp}. \quad (4.7)$$

The position in Corollary 4.1 changes exposure only along $\hat{\lambda}$. It can offset $P\zeta_J$ while leaving $\zeta_J^\perp$ unchanged. In the maintained two-shock model, complete spanning by the one claim requires

$$J_x = a(x)h_I(\alpha)J_z. \quad (4.8)$$

This is a restriction on the continuation value of worker resources relative to the claim payoff. The fact that the claim itself loads on both primitive shocks is not enough.

The distinction can be seen directly in the primitive cases from Section 2. Along the output-neutral automation direction, worker resources fall because labour loses one dollar while the capital tax recovers only a fraction; under the aligned benchmark the claim pays. Along the claim-neutral direction, worker resources still fall but the claim has zero impact payoff. No scaling of the existing position transfers an extra dollar in that state. A second claim is valuable because it rotates the payoff menu, not because it adds another source of expected return.

Proposition 4.2 (Public payoff span and its residual value). *Consider the common deterministic allocation and world prices. Suppose the value admits a regular even expansion along a selected interior optimizer, the nested asset menus stay a uniform positive distance from active constraints, and the discounted coefficients satisfy the stated integrability and terminal conditions. One claim spans the leading fiscal-resource exposure if and only if $\zeta_J^\perp = 0$. Relative to a bond-only menu, access to the maintained claim changes worker welfare by*

$$V^{\text{one}} - V^{\text{bond}} = \varepsilon^2 \int_0^\infty e^{-\rho t} \frac{\hat{\beta}_t}{2\rho} \left(\frac{s_t^0}{X_t} \right)^2 dt + O(\varepsilon^4). \quad (4.9)$$

If a second payoff completes the two-dimensional public payoff span without changing the deterministic allocation or world prices, its additional value is

$$V^{\text{complete}} - V^{\text{one}} = \varepsilon^2 \int_0^\infty e^{-\rho t} \frac{\|\zeta_{J,t}^\perp\|^2}{2\rho X_t^2} dt + O(\varepsilon^4). \quad (4.10)$$

The two welfare terms measure different margins. Equation (4.9) includes the return opportunity and the fiscal hedge available from the first claim. Equation (4.10) isolates the missing payoff direction. A completing claim may be chosen to have zero expected excess return and still deliver the second gain. That experiment holds expected-return opportunities fixed and changes only span.

The comparison completes the public payoff menu locally; it does not give workers direct market access, remove aggregate uncertainty, or produce a first best. It also presumes that the nested portfolios can be financed. At a position or solvency boundary, the KKT residual from Proposition 3.1 measures the value of additional capacity in the traded direction, while $\zeta_J^\perp$ continues to measure a different problem: exposure for which no traded direction exists.

Appendix D gives the expansion, the leading first-order condition, and the Pythagorean welfare calculation. The next section turns from state transfers to tax policy. Its central point is temporal: a predictable tax on rental flows changes the path of capital and fiscal resources, but it is not an extra payoff at the instant an automation shock arrives.

5 Taxes work on a different clock

An asset position and a capital tax act at different dates. In the Brownian economy, the inherited position in the external risky claim exposes public net worth to the next innovation. The tax rate and installed capital are instead predetermined state variables: the government can reset the tax *speed*, but not the rate or the capital stock on impact. Thus a tax cannot manufacture the missing contemporaneous payoff direction identified in the preceding section. It changes the path of the rental tax base, private accumulation, and the fiscal resources available after the innovation. Figure 3 keeps this timing distinction visible.

The distinction is literal in continuous time. Brownian paths imply that public net worth has no finite jump at date zero, $N_{0+} = N_0$ almost surely. Nevertheless, over a short interval of length Δ , the inherited risky holding supplies a payoff of order $\sqrt{\Delta}$, whereas bounded tax and other flow terms are of order Δ . The portfolio is therefore the margin that reallocates resources across the immediate innovation; tax policy is a margin for the transition that follows it. This observation does not make the portfolio an independently financed gain: under the international pricing kernel, its expected excess return compensates priced risk rather than creating free fiscal resources.

The deterministic interior characteristic makes the future-flow role of the tax explicit. Let $\mathcal{B}_u$ denote the rental-tax base, Y_u output, and $J_K(S_u)$ the marginal fiscal-resource value of an additional unit of installed capital at state S_u . Along a selected smooth interior characteristic with the stated terminal condition, the tax speed satisfies

$$\nu_t = \frac{\kappa_\tau}{Y_t} \int_t^\infty D_{t,u} \mathcal{B}_u [1 - J_K(S_u)] du. \quad (5.1)$$

Here κ_τ indexes the cost of changing the tax rate and $D_{t,u}$ is the relevant deterministic discount factor, equal to $e^{-\rho(u-t)}$ when states are frozen. The integrand is the useful economic object: a larger future base raises the return to collecting rental revenue, while a larger fiscal value of preserved capital raises the cost of reducing its after-tax return. Equation (5.1) therefore aggregates a sequence of future tax-base-capital tradeoffs; it is not an impact insurance formula.

The automation experiment shows why that distinction matters. At inherited capital, automation can lower wages and current fiscal resources even while it raises the rental tax base and the drift of capital accumulation. Thereafter, capital adjusts, and both the direct displacement effect

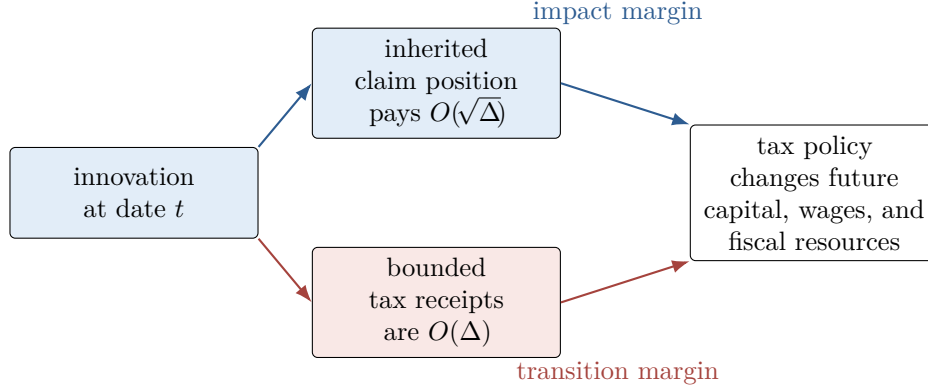


Figure 3: Impact and transition in the Brownian economy. The inherited risky holding loads public net worth on the innovation over a short interval; bounded tax receipts and capital accumulation work through subsequent flows. The figure is a timing map, not an impulse-response magnitude.

and the accumulation effect propagate with their own persistence. A tax change consequently cannot be signed from the impact tax base, the wage response, or the desired risky position alone. Its sign depends on the discounted path of $\mathcal{B}_u[1 - J_K(S_u)]$, including the terminal condition and the maintained interior conditions. When a tax, transfer, portfolio, or solvency constraint binds, the characteristic identity must be replaced by the corresponding constrained treatment.

The payoff lesson is thus narrower than “equity versus taxes.” The asset menu determines which contemporaneous Brownian directions can be transferred; tax adjustment governs a different margin, namely the future production and fiscal-resource path. The next section applies the same asset-menu question to a literal event that selects between two successor economies.

6 One arrival, two economies

The same asset-menu criterion applies when uncertainty concerns which economy follows an inaccessible automation event. This application uses a distinct small-open marked-Poisson closure: a single arrival selects either a labour-using partial-automation successor, P , or a labour-free AK successor, F . The public asset is domestic installed-capital equity, and the tax is a bounded predictable levy on rental-income flows. These are not the Brownian economy’s external claim, signed safe position, or diffusion loadings.

Hold the continuation-policy map fixed for this financial-rank comparison. Let e be public financial wealth immediately before the event and let ψ_{t-} be the predictable pre-event equity exposure. Write Δ_j^E for equity’s total-gain jump in successor $j \in \{P, F\}$, including the change in the value of installed equity. Physical capital is continuous at the event, and the pre-event equity exposure pays before any successor rebalancing. The event map is therefore

$$\begin{pmatrix} e_P^+ - e \\ e_F^+ - e \end{pmatrix} = \psi_{t-} \begin{pmatrix} \Delta_P^E \\ \Delta_F^E \end{pmatrix}. \quad (6.1)$$

The predictable rental-flow tax has no direct event atom, so it adds no second contemporaneous jump direction. Figure 4 depicts the resulting geometry.

Equation (6.1) is an exact fixed-policy rank statement for this named payoff and tax menu. If $(\Delta_P^E, \Delta_F^E) \neq (0, 0)$, one scalar exposure can move successor public wealth only along the line

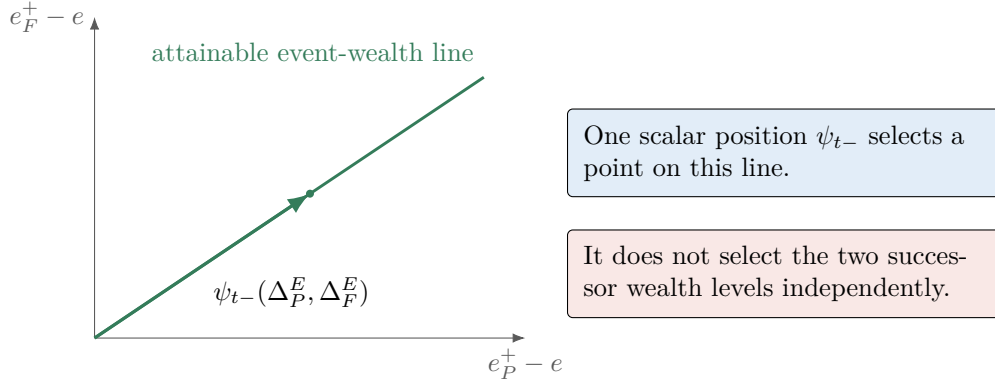


Figure 4: One event, two successor economies. With one nonzero installed-equity total-gain payoff vector, varying the scalar inherited exposure traces a line in (P, F) successor-wealth space, not the whole plane.

generated by that vector; it cannot generally choose the two successor wealth values independently. If there is only one fixed mark and its total-gain jump is nonzero, the same claim spans that one-dimensional jump-risk space. Conversely, when both successor marks are possible, a second independent direction requires a genuinely different event payoff, not simply a larger position in the same equity claim.

This is the forward-looking counterpart of the continuous-risk payoff-span argument: the missing direction is now a contingency over which successor production regime arrives. Rank alone does not establish an implementable portfolio, worker welfare, or a tax conclusion. Debt, position, transfer, protected-claim, and successor-by-successor solvency restrictions can all matter after the rank calculation. The application carries one message forward: public assets should be evaluated by the worker-resource contingencies their attainable payoffs cover, with timing and institutional closure stated for the particular economy.

7 Conclusion

Public equity is worker insurance only in the states in which it pays. Market investors price those states with one stochastic discount factor; the government values the resources they deliver to workers with another. Public trade exhausts that disagreement along the traded payoff, but cannot remove worker-resource exposure orthogonal to it.

Gross holdings therefore do not measure insurance value: they combine compensated-return demand with a fiscal hedge. The value of another claim is tied instead to the residual exposure it reaches. Capital taxation acts on a separate margin, changing future capital, wages, and the tax base. In the marked-event application, one inherited equity position likewise supplies one payoff vector across two successor economies; it does not choose worker resources separately in both.

The practical criterion is the payoff menu, not the label on the asset. Identify the states in which worker resources have high marginal value relative to market prices, then ask whether public claims pay in those states and whether the position can be financed. A different automation contingency requires a genuinely different payoff.

A Production, ownership, and equilibrium closure

This appendix fixes the maintained Brownian environment used in Sections 2–5. Workers are hand-to-mouth, domestic capitalists own installed domestic capital, international investors price the external safe claim and risky payoff, and the government chooses transfers, its net financial position, a dollar risky holding, and the speed of the capital-income tax.

Roadmap. We first state the production and ownership implications needed by the Ramsey problem. We then derive the public budget from the separate private and government accounts. Finally, we define the conditional equilibrium–Ramsey object. The load-bearing point is that wages enter worker welfare but do not become collateral for public borrowing merely because they enter worker consumption.

A.1 Production and private blocks

On the maintained full-specialization branch,

$$Y = e^{z+\nu A\alpha}\Omega(\alpha)K^\alpha L^{1-\alpha}, \quad \mathcal{B} = (R^K - \delta)K, \quad (\text{A.1})$$

where K is predetermined installed capital and $\mathcal{B}$ is the signed net-rental tax base. The competitive wage bill is $\mathcal{W} = wL$ and domestic capitalists consume $C^K = \rho K$. With $q_D = 1$, their current-state investment rule implies

$$\dot{K}/K = (1 - \tau)(R^K - \delta) - \rho. \quad (\text{A.2})$$

These equalities are maintained branch equations; they do not select that branch globally across task-regime boundaries.

A.2 Workers and the public account

Workers do not trade. Their consumption is $c = \mathcal{W} + T$, with $T \geq 0$. Thus a transfer interior has $T > 0$, whereas at $T = 0$ worker consumption is constrained by the wage. The government has financial net worth N and dollar risky holding s ; the residual safe holding is $N - s$. If the external risky total return is

$$d\mathcal{R}_R = (r + \beta_R)dt + \sigma_R^\top dW,$$

then accounting for safe return, the claim, capital-income receipts, transfers, and tax-adjustment resources gives

$$dN = [rN + s\beta_R + \mathcal{F} - c - Y\nu^2/(2\kappa_\tau)]dt + s\sigma_R^\top dW. \quad (\text{A.3})$$

Here $\mathcal{F} = \tau\mathcal{B} + \mathcal{W}$ denotes the flow of worker wages and capital-tax resources relevant for worker consumption after the private and Treasury accounts are consolidated. Wages cancel from the Treasury cash account: their inclusion in $\mathcal{F}$ is a welfare-resource accounting convention, not a pledgeability statement.

A.3 Conditional equilibrium and admissibility

Fix a state (z, x, K, N, τ) in the maintained full-specialization region. A conditional competitive equilibrium consists of the private allocations, prices, and government controls induced by that state, satisfying the private rules above, (A.3), goods feasibility, and the small-open world-price

equations. An admissible Ramsey continuation additionally satisfies the stated transfer, tax, portfolio, safe-debt, and continuation-solvency restrictions after every stopping time; it is integrable, stable under stopping-time restriction, and stable under measurable pasting. The Ramsey value is the supremum of expected discounted worker log utility over this class.

This conditional definition supports the recursion used below. It does not assert non-emptiness of the admissible class, a maximal debt frontier, or a unique global equilibrium.

B Ramsey recursion and fiscal pricing

This appendix proves the valuation statement in Proposition 3.1 on its stated smooth, transfer-interior branch.

Roadmap. The HJB supplies the portfolio Hamiltonian. Itô's formula for $e^{-\rho t}V_N$ supplies the fiscal price of the same diffusion exposure. Subtracting market and fiscal prices yields the identity; the one-dimensional KKT condition gives the bound signs. Smoothness, admissibility, and the dynamic-programming equation are maintained hypotheses throughout.

B.1 Recursive equation and pointwise controls

Let $S = (z, x, K, N, \tau)$ and suppose $V_N > 0$ and $V_{NN} < 0$ on the open region under consideration. The calculation requires that $V_N(S_t)$ is a continuous semimartingale with the displayed loading, and that the candidate s is predictable, admissible, and admits local feasible variations. Holding the state derivatives generated by (z, x, K, τ) fixed, write their contribution to the diffusion of V_N as ζ_V . The s -dependent HJB block is

$$H^s(s) = s\beta_R V_N + s\sigma_R^\top \zeta_V + \frac{1}{2}s^2 \|\sigma_R\|^2 V_{NN}. \quad (\text{B.1})$$

The transfer first-order condition is $V_N = 1/c$ when $T > 0$. The tax-speed condition, when its constraint is slack, is $\nu = \kappa_\tau V_\tau / (YV_N)$. These are necessary pointwise conditions; at an active constraint the corresponding maximization is a KKT or variational-inequality problem.

B.2 Fiscal SDF and the pricing identity

Define $m_G = e^{-\rho t}V_N(S_t)$. Itô's formula and (A.3) show that the loading of V_N in the claim direction equals

$$\sigma_R^\top \zeta_V + s \|\sigma_R\|^2 V_{NN}.$$

Consequently, defining the fiscal price of risk by the diffusion of m_G gives

$$\sigma_R^\top \lambda_G(s) = -\frac{\sigma_R^\top \zeta_V + s \|\sigma_R\|^2 V_{NN}}{V_N}. \quad (\text{B.2})$$

World no-arbitrage gives $\beta_R = \sigma_R^\top \lambda_I$. Differentiating (B.1) and substituting the two pricing relations yields

$$\frac{H_s(s)}{V_N} = \sigma_R^\top [\lambda_I - \lambda_G(s)],$$

which is Proposition 3.1's identity.

B.3 Interior and KKT conclusions

Step 1: interior stationarity. If s is interior to its feasible interval, the one-dimensional first-order condition is $H_s(s) = 0$, hence projected fiscal and market prices agree.

Step 2: bound signs. Let the feasible interval be $[\underline{s}, \bar{s}]$. At $s = \bar{s}$, feasibility permits only negative local changes, so maximization requires $H_s(\bar{s}) \geq 0$; at $s = \underline{s}$, it requires $H_s(\underline{s}) \leq 0$. Dividing by $V_N > 0$ transfers these signs to the valuation wedge. Equivalently, the multiplier on relaxing the upper bound is nonnegative; the increasing- s derivative at the lower bound is nonpositive.

Step 3: relative-consumption form. With the transfer interior and the maintained log international-investor SDF, $m_G/m_I = C^I/c$. The diffusion loading of $\log(c/C^I)$ is therefore $\lambda_G - \lambda_I$. Taking its quadratic covariation with $d\mathcal{R}_R$ gives the zero-covariance statement in Section 3 at the interior optimum.

The calculation is conditional on the dynamic-programming equation, the stated Itô regularity and integrability, admissibility, and local feasible variations. It does not verify a global candidate or characterize a production or solvency boundary.

C Fiscal resources and tax adjustment

This appendix supplies the deterministic characteristic calculation used in Section 5. It is conditional on a selected smooth interior characteristic and its terminal condition.

Roadmap. Fiscal resource value absorbs future wages and tax resources. Adding its HJB to the public budget produces comprehensive resources. Differentiating the fiscal HJB gives the tax costate, whose integrating factor gives (5.1). The decisive step is the terminal condition; without it, the identity contains an additional term.

C.1 Fiscal resources and comprehensive wealth

Let $q = (z, x, K, \tau)$, retain deterministic mean reversion through $r_0(q)$, and define $J(q)$ by its interior HJB:

$$r_0 J = \mathcal{F} + \mu_z J_z + \mu_x J_x + KgJ_K + \max_{\nu} \{J_{\tau}\nu - Y\nu^2/(2\kappa_{\tau})\}. \quad (\text{C.1})$$

Since $Y > 0$ and $\kappa_{\tau} > 0$, the expression is strictly concave in ν , yielding

$$\nu = \kappa_{\tau} J_{\tau} / Y, \quad r_0 J = \mathcal{F} + \mu_z J_z + \mu_x J_x + KgJ_K + \kappa_{\tau} J_{\tau}^2 / (2Y). \quad (\text{C.2})$$

Along the selected characteristic, (C.2) gives $\dot{J} = r_0 J - \mathcal{F} + Y\nu^2/(2\kappa_{\tau})$. Adding it to the deterministic counterpart of (A.3) gives $\dot{X} = r_0 X - c$ for $X = N + J$. At frozen common states, $r_0 = \rho$, the interior log solution is $c = \rho X$ and $V^0 = \rho^{-1} \log(\rho X)$. With deterministic mean reversion, an additive opportunity term changes value but not these N -derivative controls.

C.2 The forward-looking tax margin

Step 1: the costate equation. Set $\ell = J_K$ and $m = J_{\tau}$. Envelope differentiation of (C.1), holding the current optimizer fixed, gives

$$\dot{m} = r_0 m + \mathcal{B}(\ell - 1), \quad \nu = \kappa_{\tau} m / Y. \quad (\text{C.3})$$

The sign comes from the current gain $\mathcal{B}$ from raising the source tax versus the fiscal value J_K of the capital displaced by that tax.

Step 2: the terminal condition selects the integral. Put $D_{t,u} = \exp[-\int_t^u r_0(q_v) dv]$. Multiplying (C.3) by the integrating factor and integrating from t to T yields

$$m_t = D_{t,T}m_T + \int_t^T D_{t,u}\mathcal{B}_u(1 - J_K(q_u)) du.$$

The stipulated transversality condition $\lim_{T \rightarrow \infty} D_{t,T}m_T = 0$ and integrability of the integral imply

$$\nu_t = \frac{\kappa_\tau}{Y_t} \int_t^\infty D_{t,u}\mathcal{B}_u[1 - J_K(q_u)] du,$$

which is (5.1) at its conditional strength.

A binding tax ceiling or feasibility constraint replaces the equality by its KKT or variational-inequality counterpart. A failed terminal condition adds the displayed terminal term rather than the integral in (5.1).

D Small-risk portfolios and missing-payoff welfare

This appendix derives Corollary 4.1 and Proposition 4.2. The expansion follows a feasible separated interior branch a positive distance from the constraints.

Roadmap. Scale diffusion risk and excess returns together, then retain the order- ε^2 portfolio block. Its strict concavity gives the leading holding. Linear projection gives the exact span statement, and completing the order-two quadratic gives the two welfare terms. The remainder uses the regular even expansion and uniform interiority stated in Proposition 4.2.

D.1 Order bookkeeping and the leading first-order condition

Put $\sigma_R = \varepsilon \hat{\lambda}$ and $\beta_R = \varepsilon^2 \hat{\beta}$, where $\hat{\beta} = \hat{\lambda}^\top \hat{\lambda} > 0$. On the deterministic separated branch, $X = N + J > 0$ and

$$V_N^0 = (\rho X)^{-1}, \quad V_{NN}^0 = -(\rho X^2)^{-1}, \quad \zeta_J = (\hat{\sigma}_z J_z, \hat{\sigma}_x J_x)^\top.$$

The order- ε^2 derivative of the portfolio Hamiltonian, divided by the common positive factor V_N^0 , is

$$\hat{\beta}(1 - s/X) - \zeta_J^\top \hat{\lambda}/X.$$

The associated quadratic has second derivative $-\hat{\beta}/X^2 < 0$. Its unique interior maximizer therefore satisfies

$$s^0 = X - h_J, \quad h_J = \zeta_J^\top \hat{\lambda}/\hat{\beta}, \quad s^0 - N = J - h_J.$$

This proves the leading portfolio formula. It does not turn future worker resources into public debt capacity.

D.2 Projection and the welfare geometry

Let $P = \widehat{\lambda}\widehat{\lambda}^\top/\widehat{\beta}$. Since $P^2 = P = P^\top$, $P\zeta_J = h_J\widehat{\lambda}$ and $\zeta_J^\perp = (I - P)\zeta_J$ are orthogonal. One payoff spans the fiscal exposure if and only if $\zeta_J^\perp = 0$. In the maintained coordinates,

$$\|\zeta_J^\perp\|^2 = \frac{\widehat{\sigma}_z^2 \widehat{\sigma}_x^2 [J_x - a(x)h_I(\alpha)J_z]^2}{\widehat{\beta}}. \quad (\text{D.1})$$

Nondegenerate primitive volatilities therefore yield the body span condition.

Step 1: access to the maintained claim. Completing the strictly concave order-two portfolio quadratic relative to $s = 0$ gives the pointwise coefficient $\widehat{\beta}(s^0/X)^2/(2\rho)$. Integrating the linear order-two value equation along the common deterministic characteristic gives (4.9).

Step 2: completing the payoff menu. In a two-dimensional full-payoff menu the unconstrained exposure target is $X\widehat{\lambda} - \zeta_J$. Its Euclidean distance from the one-claim affine set is precisely $\|\zeta_J^\perp\|$ by the Pythagorean identity. The difference between the two completed squares gives (4.10).

The $O(\varepsilon^4)$ notation uses the hypotheses stated in Proposition 4.2: a regular even expansion, a smooth selected optimizer, integrable discounted coefficients, feasible nested menus, and uniform distance from active constraints. The calculation is local to that expansion and does not establish a global welfare comparison.

E Marked-event payoff span

This appendix proves the fixed-policy rank claim used in Section 6.

Roadmap. We distinguish pre-event information from post-mark continuation choices, define the total-gain payoff, and show that a predictable flow tax has no jump atom. The attainable successor-wealth map is then a one-column linear map, so its rank is at most one.

Let $\mathcal{F}_{t-}$ be pre-arrival information and let a mark $j \in \{P, F\}$ be revealed at the event. Physical capital is continuous. If aggregate installed-equity value is $Q = qK$, its proportional total-gain jump is

$$\Delta_j^E = (q_j^+ - q^-)/q^-, \quad (\text{E.1})$$

including the revaluation of the continuing installed-capital claim. A public position ψ therefore changes successor public wealth by $\psi\Delta_j^E$.

Step 1: a predictable flow tax has no event atom. The maintained tax has the form $d\mathcal{T}_t^K = \tau_t^K R_{s_t}(k_t)K_t dt$, with τ^K predictable and bounded. A finite-variation dt integral has zero mass at a totally inaccessible Poisson time. It may change continuation tax paths after the mark is observed, and hence capitalization, but it contributes no direct contemporaneous transfer vector. A total-return levy, singular jump tax, or predeclared mark-indexed instrument would be a different payoff menu.

Step 2: the event map has one column. Holding the pre-event policy and inherited balance sheet fixed, successor financial wealth satisfies

$$\begin{pmatrix} e_P^+ - e^- \\ e_F^+ - e^- \end{pmatrix} = \psi \begin{pmatrix} \Delta_P^E \\ \Delta_F^E \end{pmatrix}. \quad (\text{E.2})$$

The image is the span of one vector, hence has rank one when the total-gain vector is nonzero and rank zero otherwise. It cannot equal $\mathbb{R}^2$ for two marks. One position therefore cannot generally choose the two successor wealth values independently. With one fixed mark and a nonzero total-gain jump, the innovation space is one-dimensional and the same vector spans that jump direction.

The proof is linear algebra conditional on the stated event map. Portfolio bounds, protected inherited claims, transfers, mark-by-mark solvency, and post-event policy feasibility can prevent execution of an otherwise rank-complete direction. None is inferred from rank.

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